Yuvraj Singh Naruka

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Education

Vellore Institute of Technology, Bhopal — Bhopal, Madhya Pradesh
Bachelor of Technology in Computer Science and Engineering , CGPA: 7.51/10 — Oct 2022 – Present

Kendriya Vidyalaya No.2 Jaipur Cantt. — Jaipur, Rajasthan
Class XII (CBSE) , Percentage: 89.0% — Apr 2020 – May 2021

Kendriya Vidyalaya No.2 Jammu Cantt. — Jammu, Jammu and Kashmir
Class X (CBSE) , Percentage: 90.2% — Apr 2018 – May 2019

Technical Skills

Programming Languages: Java, Python, SQL
Web Development: HTML, CSS
Frameworks and Libraries: TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib

Technical Projects

Disaster Early Warning and Relief Resource Management System | *Python, Keras, TensorFlow* — 2026

- Directed the development of a Custom Convolutional Neural Network (CNN) to classify disaster scene images across multiple categories, achieving automated situational awareness for emergency response teams.
- Engineered and augmented a disaster image dataset through resizing, normalization, and transformations — including rotation (180°), zoom, width/height shifts, and horizontal/vertical flips — to strengthen model generalization.
- Optimized training stability with Batch Normalization, Dropout, and adaptive learning rate scheduling via ReduceLROnPlateau and EarlyStopping, using stratified train/validation/test splits for reliable evaluation.

Pneumonia Detection | *Python, Keras, TensorFlow* — 2025

- Led a team of 5 members to develop a Convolutional Neural Network (CNN) using Python, Keras, and TensorFlow to classify pneumonia from chest X-ray images, achieving 91.1% accuracy on test set.
- Processed and augmented a dataset of 5,000+ Chest X-ray images, performing resizing, normalization, and image transformations to improve model generalization.
- Boosted model performance by integrating Batch Normalization, Dropout, and learning rate scheduling, yielding high sensitivity and specificity in the final deployed model.

Heart Failure Prediction | *Python, Scikit-learn* — 2024

- Spearheaded a 5-member team to build and compare machine learning models (Random Forest, Logistic Regression, SVM) using Python and Scikit-learn, achieving 89.75% accuracy in cardiovascular risk prediction.
- Cleaned and transformed 1,000+ patient records through comprehensive data preprocessing, improving overall model reliability by 30%.
- Applied feature selection techniques and cross-validation to fine-tune model performance, with Random Forest securing the highest accuracy across all evaluated algorithms.

Loan Creditworthiness Prediction | *Python, Scikit-learn* — 2024

- Developed machine learning model (Random Forest Algorithm) using Python and Scikit-learn to predict loan creditworthiness, achieving optimal performance through RandomizedSearchCV with 5-fold cross-validation.
- Transformed loan application data across 18 features by implementing preprocessing pipelines with standardization and one-hot encoding, enhancing model reliability and input consistency.
- Refined credit risk predictions using class weighting (10:1 ratio) and probability scoring, optimizing for F1-score to improve classification performance on imbalanced datasets.

Certifications

Oracle Cloud Infrastructure Foundations Associate - *Oracle University*
Oracle Cloud Infrastructure AI Foundations Associate - *Oracle University*

Extracurricular Activities

Corresponding Author — *Infectious Diseases Now Journal* — 2025

- Authored and published editorial ”Balancing AI accuracy and medical guidance: Evaluating ChatGPT’s role in UTI-related queries” in a peer-reviewed medical journal (Volume 55, Issue 3).
- Contributed to academic discourse on AI applications in healthcare and infectious disease research, addressing ethical implications and limitations of AI-driven medical consultations.

AIEM Club — *Member* — 2024
KVS Jammu Regional Basketball Team — *Vice Captain* — 2019